

Notes2Doc: Intelligent Notes Converter

Phase 2: Agentic System Transformation

Team Members: Fatima Obaid (22I-0475), Ayesha Tahir (22I-0475), Hadia Mazhar (22I-0487)

Course: Professional Practices in IT (PPIT)

2. Phase 1 Recap

- Problem: Students struggle to digitize handwritten notes.

- Users: S

3. Computing as a Formal Profession

- Software development is a formal profession requiring standards.

- Respons

4. Ethics vs. Morals

- Morals: Personal beliefs (e.g., education should be free).

- Ethics: P

5. Professional Ethics

- Code Quality: Robust failovers instead of crashes.

- Security:

6. Ethical Decision Making

- Dilemma: Speed vs. Quality.

- Decision
complex/p

7. Importance of Ethical Decision Making

- On Users: Reliable transcriptions prevent mistakes.

- On Socie

8. Ethical Theories & Human-Centered Design

- Applied Theory: Deontology (Duty-based ethics).

- Duty to r

9. ACM / IEEE Code of Ethics

- Followed (ACM 1.2): Deleting data immediately to avoid leaks.

- Followed

10. Ethical Decision Process (4-Step Model)

- Issue: Agent needs to learn, but must protect privacy.

- Analyze:

11. Software Development & Industry Practices

- Workflow: Modular, component-based architecture in Phase 2.

- Version C

12. Trends in IT & Agentic Systems

- Industry shift from static AI to Autonomous Agents.

- Notes2D

13. Career Relevance

- Skills: Prompt engineering, Agentic architectures, Python.

- Ethics: U

14. Virtual Work & Sustainability

- Remote Collaboration: Cloud IDEs and version control.

- Green C
footprint.

15. Legal Aspects of Computing

- Data Protection: Securing API keys and user uploads.

- User Rig

16. Intellectual Property Rights (IPR)

- App Ownership: Code owned by the student team.

- Licensing

17. Computer Crimes & Risks

- Data Theft: Mitigated by in-memory processing.

- Misuse: I

18. Computer Contracts

- Terms of Service: Privacy & Consent Notice acts as a micro-contract.

- Develop

19. Technical Limitations of Phase 1

- Static Logic: Hardcoded prompt for every image.

- No Auton

20. Agentic System Concept

- Perception: Understanding image quality.

- Decision

21. Gap Analysis

- Phase 1 lacked perception, decision engines, and memory.

22. Agentic Vision

- Tool -> Agent

- Reactive

23. Agent Architecture

- Flow: Input -> Perception -> Decision -> Action -> Feedback Loop.

- Memory:

24. Agent Type Selection

- Type: Goal-based / Learning Agent.

- Goal: Ma

25. Operational Workflow

- Observe: Analyze brightness, contrast, content type.

- Interpret:

26. Intelligence Layer

- Rules/Heuristics for Perception & Decision.

- LLMs (G

27. Memory & Context

- Short-term Memory: Session thought log, decisions, confidence.

- Long-term

28. Autonomy Level

- Semi-Autonomous.

- Agent ha

29. Human-in-the-Loop

- Initial image upload and consent.

- Review/e

30. Ethical Agent Design

- Privacy: No image persistence.

- Bias: Ad

31. Risk Assessment

- Incorrect Decisions: Misinterpreting diagrams as text.

- Over-aut

32. Safety Mechanisms

- Logging: Full audit trail.

- Override

Comparative Analysis

- Control: User-driven (Phase 1) vs System-driven (Phase 2)

- Intelligent